

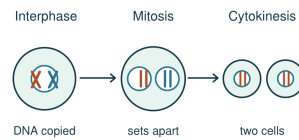


I am learning to order the cell cycle and explain why a cell makes two matching cells.

Words to use

Word	Meaning and example	My language
cell cycle	the series of steps in which a cell grows and divides The cell cycle includes interphase, mitosis and cytokinesis.	_____
interphase	the stage in which the cell grows and copies its DNA The cell copies its DNA in interphase.	_____
mitosis	the stage in which copied chromosomes separate into two nuclei Mitosis separates the chromosome sets.	_____
chromosome	a structure of DNA that carries genes Each new nucleus receives a matching set of chromosomes.	_____
cytokinesis	the division of the cell after the nuclei form Cytokinesis splits one cell into two.	_____
prophase	the mitosis phase when chromosomes condense In prophase, chromosomes become easier to see.	_____
metaphase	the mitosis phase when chromosomes line up in the middle In metaphase, chromosomes form a line.	_____
anaphase	the mitosis phase when the two copies of each chromosome are pulled to opposite ends In anaphase, the two copies of each chromosome are pulled to opposite ends.	_____
telophase	the mitosis phase when two new nuclei form In telophase, each set gains a nucleus.	_____

Model



Body-cell division: copied chromosomes separate, then one cell splits into two. Simplified; not to scale.

Interphase: copy DNA. Mitosis: prophase (condense), metaphase (line up), anaphase (the two copies of each chromosome are pulled to opposite ends), telophase (two nuclei). Cytokinesis: split the cell.

How I know

I can order interphase, four mitosis phases and cytokinesis.

I can show that the two new body cells have matching chromosome sets.

I can explain how mitosis helps growth or repair.

You can plan in any language. Speak before you write: The daughter cells match because ...